Thomas Randall, PhD Computer Science

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OBJECTIVE:

<OPTIONAL: 1 sentence about the specific position/company (name it) and your interest/knowledge that makes you most relevant>

EDUCATION:

Doctor of Philosophy in Computer Science — May 2025
Clemson University — Clemson, SC

Bachelor of Science in Computer Science with Minor in Business Administration — May 2019
Clemson University — Clemson, SC

RELEVANT EXPERIENCE:

EXPERIENCE:

Volunteer Research Assistant — Fall 2025—Present
Clemson University — Clemson, SC

- Fostered cross-laboratory collaborations within Clemson University, leading to a scientific publication (TBA)

Graduate Research Assistant — Summer 2019—Spring 2025
Clemson University — Clemson, SC

- Analyzed thermal behaviors for High Performance Computing clusters in Single Phase Liquid Immersion Cooling environments with novel data harness
- Optimized the Word2Vec Neural Network for NVIDIA Pascal and Volta architectures, resulting in 472% higher throughput while improving between-architecture scaling by 197%
- Established mutually beneficial collaborations between Clemson University and Iowa State University, as well as two Department of Energy National Laboratories (Argonne and Oak Ridge) leading to multiple paper publications
- Mentored other graduate students in scientific experimentation, paper writing, and HPC optimization, contributing to their masters' and doctoral research objectives
- Presented scientific breakthroughs and research updates in weekly meetings
- Published three scientific papers in top-tier conferences and additional workshop papers, including a Best Paper Award in the International Conference on Supercomputing 2021

Vice President — Fall 2024—Spring 2025
School of Computing Graduate Student Association — Clemson, SC

- Organized 3 professional workshops, 3 social events and a fundraiser, increasing student engagement
- Managed organization finances, plans, marketing and personnel to revitalize organization post-COVID

OMNI Internship Program, Graduate Student Researcher — Summer 2024
Oak Ridge National Laboratory — Oak Ridge, TN

- Identified key insights on LLM in-context learning and inference behaviors that limit current capabilities to generalize across new data without additional training or fine-tuning
- Mentored and supervised two other students in program, helping one establish themselves with their first scientific publication
- Published work in the 2025 HPC for AI Foundation Models and LLMs for Science (HPAI4S)

Lead Student Volunteer, Guided Interest Group Leader, HPC Immersion Mentor — Fall 2023, 2024
SuperComputing '23, '24 — Denver, CO; Atlanta, GA

- Established Reproducibility efforts and ACM Badge Verification processes for the SuperComputing conference
- Organized over 30 conference Panels and Birds of a Feather with conference committee members
- Managed 100+ student volunteers to facilitate conference events and coordinated efforts serving up to 15,000 guests
- Developed conference attendance plans and paper discussion sessions for groups of three to seven students, focusing on encouraging and empowering students to make the most of their conference attendance while fostering a learning environment
- Formed individual mentoring relationships in one-on-one settings with newcomers to the High Performance Computing field, ensuring they were welcomed and involved in their conference experience

GIVENS Scholar, Graduate Student Researcher — Summer 2020, 2022, 2023
Argonne National Laboratory — Lemont, IL

- Created a novel methodology for automatic performance tuning via transfer learning with probabilistic performance modeling that achieves 33.39x additional speedup over prior state-of-the-art techniques
- Published work in the International Conference on Supercomputing (ICS), a top-tier peer-reviewed computer science conference
- Assisted in mentoring other students with basic use of High Performance Computing systems

Graduate Teaching Assistant — Fall 2019 — Spring 2023
Clemson University, School of Computing — Clemson, SC

- Prepared assignments, projects and homework for classes with 30-100+ students (up to 8 hours per week)
- Aided students in office hours with questions about course materials and assignments (3-5 hours per week)

- Developed a new course for Clemson University with Assistant Professor Dr. Mert Pesé to enrich curriculum for the newly formed Computer Science concentration in Cybersecurity
- Delivered consistent and actionable feedback to students within 7 days of submission on labs, projects, quizzes and exams

Student Volunteer, Virtual Student Volunteer Fall 2020 (Virtual), Fall 2021, Fall 2022
SuperComputing'20, SuperComputing'21, SuperComputing'22 Online; St. Louis, MO; Dallas, TX
- Moderated online discussions and forward questions for panels and technical presenters with over 1000 live members in the audience
- Facilitated chain-of-command communications between conference organizers, volunteers, building staff and event security, improving response times to emergency situations

Undergraduate Research Assistant Summer 2018—Spring 2019
Clemson University Clemson, SC
- Optimized a CPU kernel to accelerate Word2Vec kernel performance, achieving 1.3x speedup over prior state-of-the-art throughput
- Performed GPU kernel performance analysis and development, identifying key bottlenecks and possible algorithmic remedies to address performance scaling issues
- Assimilated in-depth background research and algorithmic analysis of full-scale machine learning application
- Created and delivered presentations to demonstrate learning progress and setbacks for weekly group presentations

Laboratory Teaching Assistant Spring 2017 — Spring 2019
Clemson University, School of Computing Clemson, SC
- Prepared and introduced labs and projects for up to 30 students (up to 8 hours per week)
- Delivered consistent and actionable feedback to students with 7-day turnarounds on submitted labs and projects
- Aided students with questions and problems during lab time and office hours (3-5 hours per week)

Summer Staff Coordinator Summer 2017
Isaiah 55 Deaf Ministries La Feria, TX
- Coordinated and supervised volunteer teams of 5-15 on small-scale construction and renovation projects, completing 30 projects over a 10-week summer term
- Organized daily community outreach events including VBS and children's arts & crafts and recreation, serving 10-100 children per day
- Managed logistics for tool distribution and construction materials to reduce downtime setting up and closing down workdays

SKILLS:

ENGAGEMENT (Communication, Collaboration, Leadership)
- Scientific writing
- Conference presentations (public speaking)
- Mentoring graduate and undergraduate students

INNOVATION (Adapatability, Analysis, Technology)
- Scientific research and analysis synthesis
- Statistical performance modeling (machine-learned)
- Performance optimization
- High Performance Computing
- C, CUDA, Python

AWARDS:

Outstanding PhD Student in Computer Science Spring 2025

ICS'21 Best Paper Summer 2021

R.C. Edwards Graduate Fellow Fall 2019—Spring 2022

Clemson School of Computing Outstanding Undergraduate Researcher Spring 2019

Dupont Best Undergraduate Project Spring 2019

INVITED TALKS AND FEATURED COVERAGE:

Oak Ridge National Lab Artificial Intelligence Seminar Series Spring 2024
Invited Presenter Oak Ridge, TN

Argonne National Lab Computer Science Seminar Series Summer 2023
Invited Presenter Lemont, IL

I Am HPC Spring 2023
https://sc23.supercomputing.org/2023/05/hpc-on-the-rise-thomas-randall/ Online

Ask A Grad Fall 2022
SoC-GSA & UPE Clemson, SC